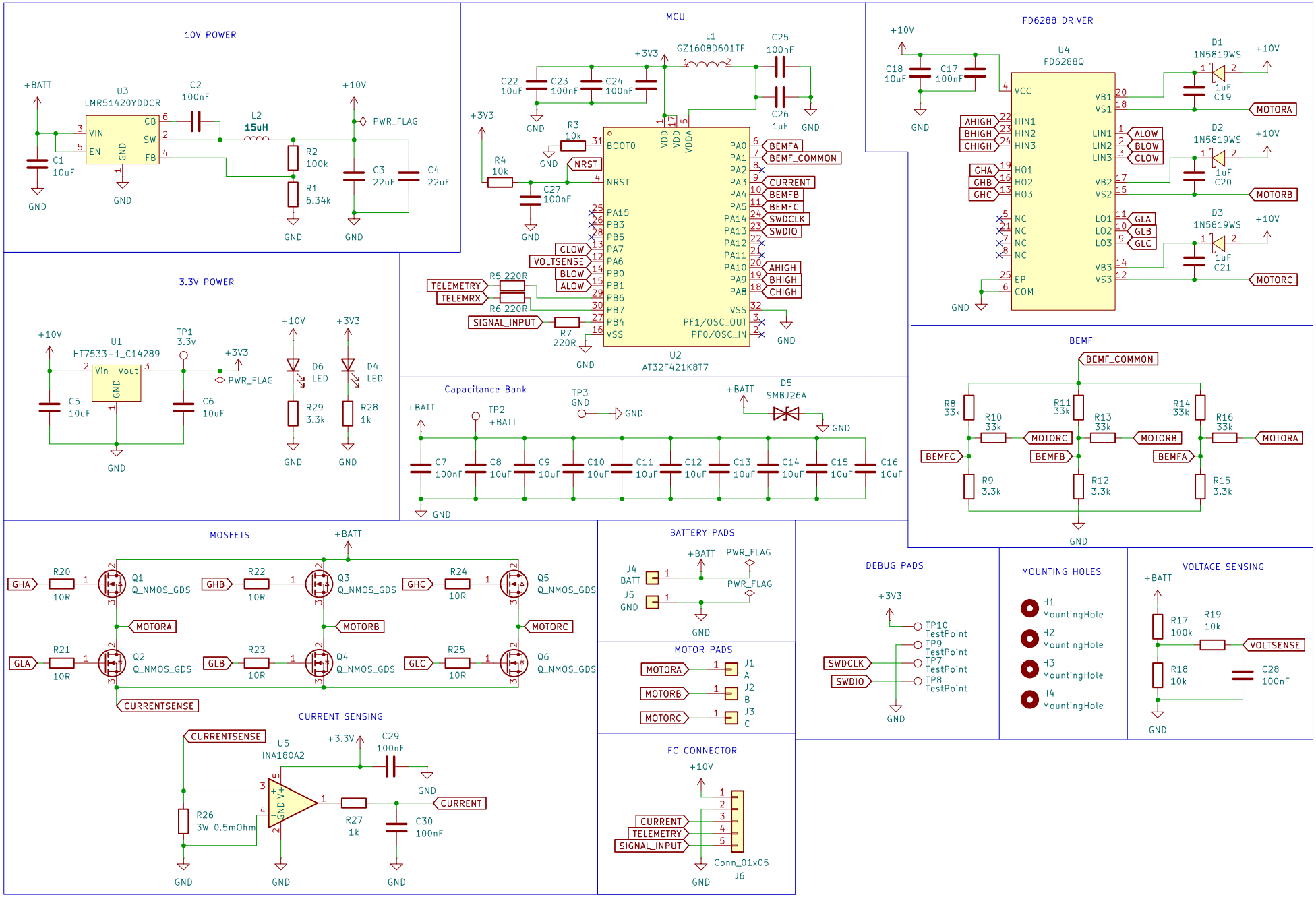
10V POWER
+BATT
U3
LMR51420YDDCR
VIN
EN
GND
CB
SW
FB
C1
10uF
C2
100nF
L2
15uH
R2
100k
R1
6.34k
C3
22uF
C4
22uF
+10V
PWR_FLAG
3.3V POWER
U1
HT7533-1_C14289
Vin
Vout
TP1
3.3v
C5
10uF
C6
10uF
PWR_FLAG
D6
LED
R29
3.3k
D4
LED
R28
1k
MCU
L1
GZ1608D601TF
+3V3
C22
10uF
C23
100nF
C24
100nF
C25
100nF
C26
1uF
R3
10k
R4
10k
NRST
C27
100nF
BOOT0
VDD
VDDA
PA15
PB3
PB5
PA7
PA6
PB0
PB1
PB6
PB7
PB4
VSS
PA0
PA1
PA2
PA3
PA4
PA5
PA14
PA13
PA12
PA11
PA10
PA9
PA8
PF1/OSC_OUT
PF0/OSC_IN
BEMFA
BEMF_COMMON
CURRENT
BEMFB
BEMFC
SWDCLK
SWDIO
CLOW
VOLTSENSE
BLOW
ALOW
AHIGH
BHIGH
CHIGH
R5 220R
R6 220R
R7
220R
TELEMETRY
TELEMRX
SIGNAL_INPUT
U2
AT32F421K8T7
FD6288 DRIVER
U4
FD6288Q
C18
10uF
C17
100nF
VCC
HIN1
HIN2
HIN3
GHA
GHB
GHC
NC
EP
COM
VB1
VS1
LIN1
LIN2
LIN3
HO1
HO2
HO3
VB2
VS2
LO1
LO2
LO3
VB3
VS3
D1
1N5819WS
D2
1N5819WS
D3
1N5819WS
C19
1uF
C20
1uF
C21
1uF
MOTORA
MOTORB
MOTORC
Capacitance Bank
TP2
+BATT
TP3
GND
D5
SMBJ26A
C7
100nF
C8
10uF
C9
10uF
C10
10uF
C11
10uF
C12
10uF
C13
10uF
C14
10uF
C15
10uF
C16
10uF
BEMF
BEMF_COMMON
R8
33k
R10
33k
R11
33k
R13
33k
R14
33k
R16
33k
R9
3.3k
R12
3.3k
R15
3.3k
MOSFETS
GHA
GLA
GHB
GLB
GHC
GLC
R20
10R
R21
10R
R22
10R
R23
10R
R24
10R
R25
10R
Q1
Q_NMOS_GDS
Q2
Q_NMOS_GDS
Q3
Q_NMOS_GDS
Q4
Q_NMOS_GDS
Q5
Q_NMOS_GDS
Q6
Q_NMOS_GDS
CURRENTSENSE
BATTERY PADS
J4
BATT
J5
GND
DEBUG PADS
SWDCLK
SWDIO
TP10
TestPoint
TP9
TestPoint
TP7
TestPoint
TP8
TestPoint
MOUNTING HOLES
H1
MountingHole
H2
MountingHole
H3
MountingHole
H4
MountingHole
VOLTAGE SENSING
R17
100k
R19
10k
R18
10k
C28
100nF
VOLTSENSE
MOTOR PADS
J1
A
J2
B
J3
C
CURRENT SENSING
U5
INA180A2
C29
100nF
C30
100nF
R26
3W 0.5mOhm
R27
1k
CURRENT
FC CONNECTOR
TELEMETRY
SIGNAL_INPUT
Conn_01x05
J6